

of an object along the screen. This is mainly because between the initialisation of the image on the first frame and the second frame, the image is stuck and held on the first frame. This is the judder effect. Smearing is similar to strobing and motion blur, that is, when the image is retained on a previous pixel even after it has been re-rendered on another pixel, partly due to slow refresh rates and partly due to the brain's retentivity.

Absentia VR, an Indian VR startup, has used a technique called Black Frame Insertion to deal with this in their VR headset, the Tesseract. This technique has been around with televisions for a while and is currently being adopted by multiple VR manufacturers. The approach is to insert a black frame between each frame of content, causing both the brain and the display to initialize the display after each frame, understandably nullifying both judder and smear. The device also uses some other software mechanisms to make the VR experience as less jarring as possible, like false nose simulation, and using an patent-pending algorithm developed by them to turn any 2D content to stereoscopic 3D by intelligent depth perception. The Tesseract also uses smart vectorization to define the edges and depth of an object in real time through its algorithm, which further eliminates any drift that it might have suffered otherwise. To know more about the Tesseract go here: <http://dgit.in/AbsentiaVR>

Where is all the VR?

Honestly, the VR buzz has been around for a while. Although most of the devices are yet to reach the hands of their first users, Virtual reality aren't a lot more accessible than say a couple of years ago. Recently, the Oculus Rift launched at a retail price of \$599, or roughly INR40,000. Needless to say, this makes it more of a luxury gadget than something someone would make an integral part of their life, especially due to the current lack of specialised content. Many other high-grade VR HMDs are going in the same direction and the general public is only experiencing them at Technology exhibitions and Industrial conferences.

On the other hand, Google Cardboard has aimed to make VR affordable and accessible to all with open source designs. But while using the Cardboard, the experience is not even close to being as real as with the previous headsets. Even with several Cardboard inspired designs as well as Samsung's GearVR, many of the problems discussed in this chapter are glaringly apparent.

To solve this problem, a sweet middle spot needs to be struck. And to do that, we might need to take up things into our own hands. That is, if it was